# Rithvik Mani

Champaign, IL

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## Education

- **Bachelors of Science in Bioengineering + Computer Science** — *Aug 2022 - May 2026*
  *University of Illinois at Urbana-Champaign, Grainger College of Engineering* — GPA: 3.5
  - **Courses**: Data Structures & Algorithms, Systems Programming (C), Machine Learning, Biomedical Imaging & Deep Learning, Linear Algebra, Systems and Signals, Distributed Systems, Bioinformatics Algorithms

## Experience

- **US Department of Agriculture** — *May 2025 - Aug 2025*
  *Artificial Intelligence (AI) and Robotics Research Intern* — Gainesville, FL
  - Engineered a real-time **computer vision pipeline** leveraging **Gaussian Splatting**, **MVS**, and **SFM** to generate high-fidelity 3D reconstructions of agricultural environments, enabling precise field phenotyping and robot navigation.
  - Optimized GPU-accelerated inference using **CUDA**, achieving **3x speedup** in object detection and mapping tasks while integrating **VGGT** for state-of-the-art visual grounding and minimizing memory bottlenecks.
  - Developed autonomous data-collection workflows with **ROS2** and **ZED SDK**, orchestrating multi-camera stereo vision, 3D segmentation with various models such as **Open3D** and **Segment Anything Model**.
  - Built scalable codebases in **Python** and **C++**, containerized workflows with **Docker**, and automated data preprocessing pipelines, and accelerating downstream bioinformatics analysis and ML model retraining.
- **Arcel AI (arcel.ai)** — *April 2025 - present*
  *Co-Founder* — Champaign, IL
  - Co-founded **Arcel AI**, an AI-powered spreadsheet copilot built for FP&A and accounting pros that integrates directly with Google Sheets and MS Excel to be an agentic command center for the entire financial stack. Designed and led scalable backend architecture serving while maintaining sub-**200ms latency** for a seamless user experience.
  - Spearheaded development of intelligent cell-edit tracking and visualization, introducing **color-coded diffing** (additions, modifications, removals) and real-time change reconciliation to improve transparency and user trust.
  - Built production services using **React**, **Next.js**, **Python/FastAPI**, **PostgreSQL**, and **Redis**; deployed on **render.com**
- **US Army Corps of Engineers** — *Jun 2024 - Aug 2024*
  *Computational and Synthetic Engineering Biology Intern* — Champaign, IL
  - Engineered **computational models** of **B. subtilis spore germination** kinetics, simulating nutrient-triggered germination pathways and PCR to predict germination rates and heterogeneity under diverse environmental conditions.
  - Built automated **data preprocessing and analysis pipelines** in **Python** to process germination assay outputs.
- **Equilocalm, Carle Illinois College of Medicine** — *May 2024 - Aug 2024*
  *Research and Development Intern* — Champaign, IL
  - Contributed to the development of MenoPatch, a trans-dermal skin patch to treat menopause symptoms.
  - Researched and tested different types of hydrogels in vitro and implemented them in the prototype.
  - Assisted in optimizing prototype for 3D printing production through heat sealing and selection of materials.
- **Syneos Health** — *May 2023 - Aug 2024*
  *Medical Value & Access Consulting Intern* — New York, NY
  - Collaborated with pharma clients in performing market access viability & compliance determination for lead drugs
  - Interviewed insurance professionals, developed Discussion Guide, analyzed Non-Alcoholic Steatohepatitis TPP
- **Founders - Illinois Entrepreneurs** — *Jan 2023 - Jan 2025*
  *Executive Board, Head of Events* — Champaign, IL
  - Student led 501(c)(3) organization focused on promoting and fostering entrepreneurship throughout the Midwest
  - Led the Events team, organized startup events, brought in over 80 startups and VCs, gained $10,000 in sponsorships

## Projects

- **Protein Binder Generator** — *AWS Cloud, Docker, Python, Terraform, AWS CloudFormation, AI, ML*
  - Built end-to-end pipeline for AI protein binder design on **AWS** using **RFDiffusion, ProteinMPNN, and ColabFold**
  - Serverless workflow using **AWS Batch, Lambda, DynamoDB, and S3.** Automates binder generation from target proteins & hotspots. Containerized pipeline for backbone generation, sequence design, and structure validation.
  - Designed event-driven architecture where uploading target file to S3 automatically triggered the full binder design workflow. Accepts PDB Files or FASTA sequences with the ability to generate up to 10 designs per binding.
- **Stethopy** — *C++, React Native, Python, Node.js, Docker*
  - Developed mobile application and medical device kit for telehealth doctors for stethoscope use over the phone
  - Coded back-end functionality with C++, coded app UI with React Native, and cleaned .wav files using Python

## Technical Skills

**Languages**: Python, C/C++, Java, SQL, JavaScript, TypeScript, Bash, MATLAB, R, HTML+CSS
**Frameworks & Libraries**: PyTorch, CUDA, NumPy, Pandas, Scikit-learn, LangChain, FastAPI, React, Next.js, ROS2
**Developer Tools**: Git/GitHub, Docker, Kubernetes, CI/CD (GitHub Actions), CMake, GDB, UNIX/Linux, ZED SDK
**Cloud/DB**: GCP (Cloud Run, GKE, Firestore), AWS (DynamoDB, S3, Batch), PostgreSQL, MySQL, Redis, Supabase
**Topics**: Computer Vision (SLAM, MVS, SFM), LLMs, HPC/Parallel Computing, Bioinformatics Pipelines